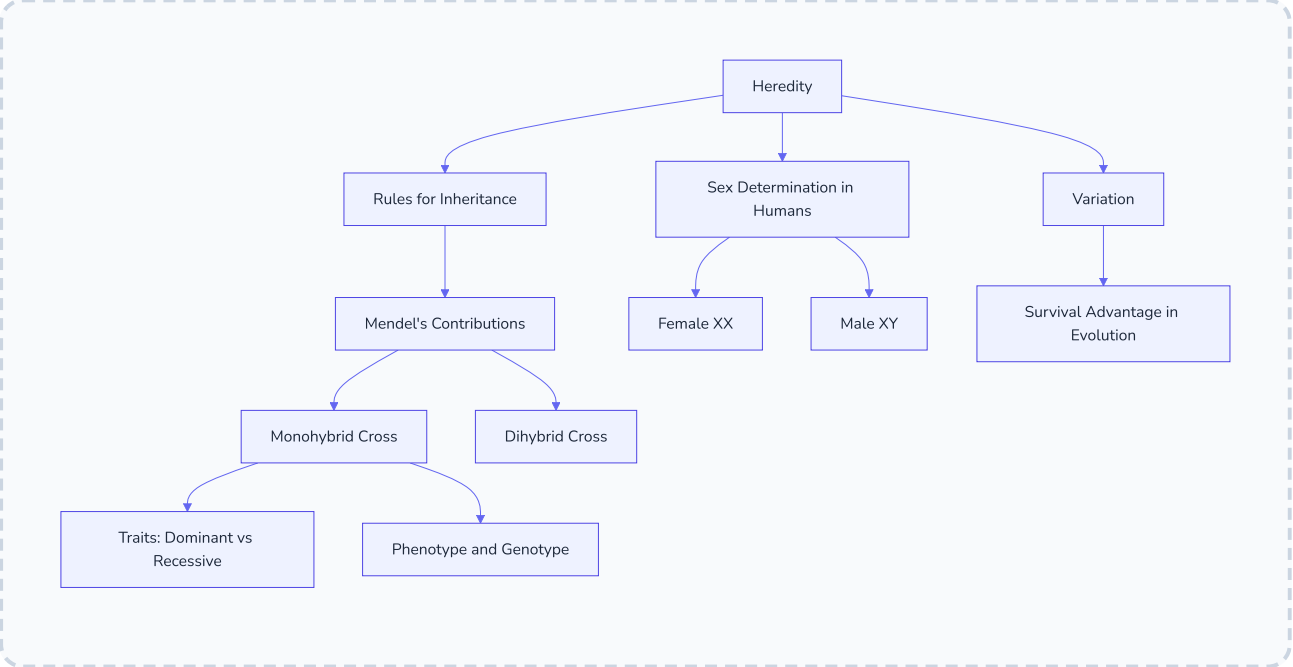
Chapter 8: Heredity 🧬 👨‍👩‍👧

Have you ever wondered why you look a little bit like your mother and a little bit like your father? Why do siblings raised in the exact same environment still look different? This happens because of **Heredity and Variation**.

Master Mind Map 🗺️



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1. Accumulation of Variation During Reproduction 📈

- **Heredity:** The transmission of characters (traits) from parents to their offspring.
- **Variation:** The differences in the characters among the individuals of a species.

During reproduction, DNA is copied. No copying mechanism is 100% perfect.

- **In Asexual Reproduction**: Very minor variations exist (due to small DNA copying inaccuracies).
- **In Sexual Reproduction**: Large variations exist because DNA from two different parents combine.

> **QUICK TIP 💡**
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> Depending on the nature of variations, different individuals would have different kinds of advantages. Bacteria that can withstand heat will survive better in a heat wave.

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2. Rules for the Inheritance of Traits: Mendel's Contributions 🌱

Gregor Johann Mendel (The Father of Genetics) used a number of contrasting visible characters of garden peas (*Pisum sativum*) like round/wrinkled seeds, tall/short plants, white/violet flowers to study inheritance.

Mendel Selected Pea Plants Because:

1. They have clearly contrasting characters (e.g., tall vs. short).
2. They have a short life span (can study many generations quickly).
3. They normally self-pollinate, but cross-pollination can be easily done.

Key Terms in Genetics

Term	Definition
Gene	The functional unit of heredity found on chromosomes. Controls a specific trait.
Allele	Alternate forms of a gene (e.g., 'T' for tall, 't' for dwarf).

Dominant Trait	The trait that expresses itself in the presence of its contrasting allele (represented by capital letters like 'T').
Recessive Trait	The trait that remains suppressed and only expresses itself when paired with identical alleles (represented by small letters like 't').
Genotype	The genetic makeup of an organism (e.g., TT, Tt, tt).
Phenotype	The observable physical appearance (e.g., Tall, Dwarf).

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3. Monohybrid Cross (Inheritance of One Trait) 🌿

A cross between two individuals taking only ONE contrasting character into consideration.
Example: Crossing a Pure Tall (TT) plant with a Pure Dwarf (tt) plant.

First Generation (F1)

- Parents: TT $\times$ tt
- F1 offspring: All plants were **Tt** (Tall).
- *Conclusion:* The trait that appeared in F1 is dominant (Tallness). The one that did not appear is recessive (Dwarfness).

Second Generation (F2) Mendel self-pollinated the F1 plants (Tt $\times$ Tt).

- F2 Offspring: 3 Tall plants : 1 Dwarf plant.
- **Phenotypic Ratio:** 3:1 (Tall : Dwarf)
- **Genotypic Ratio:** 1:2:1 (TT : Tt : tt)

IMPORTANT POINTS 📌

The F2 generation showed that the 't' (dwarf) trait had not vanished; it was merely hidden in the F1 generation and reappeared when two 't' alleles paired up.

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4. Dihybrid Cross (Inheritance of Two Traits) 🌾

A cross between two plants having TWO pairs of contrasting characters. *Example: Seed shape (Round/Wrinkled) AND Seed color (Yellow/Green).*

- Round (R) is dominant over Wrinkled (r).
- Yellow (Y) is dominant over Green (y).

Cross: Round Yellow (RRYY) $\times$ Wrinkled Green (rryy)

- **F1 Generation:** All plants produced Round Yellow seeds (RrYy).
- **F2 Generation (Self-pollination of F1):** Produced 4 different types of seeds.
- **Phenotypic Ratio in F2:** 9 : 3 : 3 : 1 (9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green)

> **NOTE** 📝
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> **Law of Independent Assortment**: The appearance of new combinations (Round Green and Wrinkled Yellow) proved that traits are inherited independently of each other. The gene for shape does not control or affect the gene for color.

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5. How do these Traits get Expressed? ⚙️

Cellular DNA is the information source for making proteins in the cell. A section of DNA that provides information for one protein is called the **Gene** for that protein.

- *Example:* Plant height depends on the amount of a particular growth hormone. The amount of hormone depends on the efficiency of the enzyme (protein) that synthesizes it.
- If the gene is efficient/normal (Dominant 'T'), a lot of hormone is made $\rightarrow$ Plant is Tall.

- If the gene is altered (Recessive 't'), less hormone is making $\rightarrow$ Plant is Short.

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6. Sex Determination in Humans 👫

Different species use different strategies for sex determination. In humans, sex is determined genetically by chromosomes.

- Humans have **23 pairs of chromosomes** in each cell.
- 22 pairs are perfectly matched (Autosomes).
- The 23rd pair are the **Sex Chromosomes**, denoted as X and Y.
  - Females have a perfect pair: **XX**
  - Males have a mismatched pair: **XY**

Mechanism:

- All children inherit an **X chromosome** from their mother, regardless of gender.
- The sex of the child depends completely on what they inherit from the father.
- If the sperm carries an **X chromosome**, the child will be a **girl (XX)**.
- If the sperm carries a **Y chromosome**, the child will be a **boy (XY)**.

> ⚠️ **COMMON MISTAKES** ❌
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> There is exactly a **50% probability** of a child being a boy or a girl. It is scientifically incorrect to blame the mother for the sex of the child, as the deciding chromosome (Y) exclusively comes from the father.

Quick Revision Notes 📌

1. **Heredity**: Traits passed from parents to offspring.
2. **Mendel's Pea Plants**: Ideal subjects due to distinct contrasting traits, short lifespans, and ease of cross-pollination.

3. **Monohybrid Ratio**: 3:1 (Physical appearance) / 1:2:1 (Genetic makeup).
4. **Dihybrid Ratio**: 9:3:3:1
5. **Traits expression**: Genes control proteins $\rightarrow$ Proteins control hormones $\rightarrow$ Hormones control traits.
6. **Human Genetics**: Females are XX. Males are XY. Fathers determine the sex of the child.

Chapter Summary 🏁

Heredity explains how biological traits are passed from one generation to the next. Gregor Mendel laid the foundation of genetics by crossing pea plants and discovering that traits exist as pairs of alleles (dominant or recessive). He proved traits segregate and assort independently of each other. At the cellular level, genes hold the DNA blueprints for proteins that physically express these traits. In humans, our sex chromosomes (XX or XY) dictate biological sex, with the biological father's sperm carrying the deciding X or Y chromosome.